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| **Status** | Finished |
| **Started** | Wednesday, 28 October 2026, 7:48 PM |
| **Completed** | Wednesday, 28 October 2026, 7:53 PM |
| **Duration** | 5 mins 11 secs |
| **Grade** | **13.00** out of 15.00 (**86.67**%) |

Question **1**
Correct
Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It can only be applied to large lists

Question **2**
Correct
Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓
b. When the item is not the array at all
c. When the item is the last element in the array
d. Item is the last element in the array or item is not there at all

Question **3**
Correct
Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The first element is returned
c. The last element is returned
d. The search is considered unsuccessful ✓

Question **4**
Correct
Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O👎 ✓
c. O(1)
d. O(log n)

Question **5**
Correct
Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. At the beginning of the list
- c. In the left sublist
- d. In the right sublist ✓

Question **6**
Correct
Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Must use a sorted array
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- d. There must be a mechanism to access middle element directly

Question **7**
Correct
Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Both (1) & (3)
- b. Linear
- c. Hash
- d. Binary ✓

Question **8**
Correct
Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Hashing
- c. Linear search ✓
- d. Dynamic search

Question **9**
Incorrect
Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the middle element in the list
- b. Sort the list ✗
- c. Compare the target element with the first element in the list
- d. Divide the list into two equal parts

Question **10**
Correct
Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By using a hash function
- c. By sorting the list first
- d. By dividing the list into halves

Question **11**
Correct
Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Binary search
- b. Both (1) & (3)
- c. Linear search ✓
- d. Hash search

Question **12**
Incorrect
Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 94 ✗
- b. 89 and 99
- c. 89 and 94
- d. 90 and 99

Question **13**
Correct
Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and sorted
- b. When the list is large and sorted
- c. When the list is large and unsorted
- d. When the list is small and unsorted ✓

Question **14**
Correct
Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n-1
- b. log n
- c. n/2
- d. n ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the right sublist
c. In the entire list
d. In the left sublist ✓